1. A particle moves with velocity $v = kt^2$. The acceleration is

A) kt

B) 2kt

C) t^2

D) 2k

Answer: B

---

2. A body is projected vertically upward with velocity u. Time to reach maximum height is

A) u/g

B) u/2g

C) 2u/g

D) g/u

Answer: A

---

3. Work done by a conservative force in a closed path is

A) Maximum

B) Minimum

C) Zero

D) Infinite

Answer: C

---

4. If kinetic energy is doubled, momentum becomes

A) √2 times

B) 2 times

C) 4 times

D) 1/2 times

Answer: A

---

5. The dimensional formula of force is

A) MLT^{-2}

B) ML^2T^{-2}

C) MLT^{-1}

D) $ML^{-1}T^{-2}$

Answer: A

---

6. Escape velocity depends on

A) Mass of planet

B) Radius of planet

C) Both A and B

D) None

Answer: C

---

7. A satellite orbiting close to Earth has speed

A) 11.2 km/s

B) 7.9 km/s

C) 3 km/s

D) 1 km/s

Answer: B

---

8. The unit of angular momentum is

A) kg m^2/s

B) kg m/s^2

C) N m

D) J

Answer: A

---

9. Torque is equal to

A) $r \times F$

B) rF

C) F/r

D) r^2F

Answer: A

10. The moment of inertia depends on

A) Mass

B) Distribution of mass

C) Axis of rotation

D) All of these

Answer: D

---

11. A body in uniform circular motion has

A) Zero acceleration

B) Constant acceleration

C) Variable acceleration

D) Infinite acceleration

Answer: C

---

12. Centripetal force acts

A) Away from center

B) Towards center

C) Tangentially

D) Upward

Answer: B

---

13. The unit of surface tension is

A) N/m

B) N

C) J

D) Pa

Answer: A

---

14. Pressure in a liquid increases with

A) Depth

B) Area

C) Volume

D) Density

Answer: A

---

15. Bernoulli's principle is based on conservation of

A) Energy

B) Mass

C) Momentum

D) Charge

Answer: A

---

16. Young's modulus is

A) Stress/Strain

B) Strain/Stress

C) Force/Area

D) Area/Force

Answer: A

---

17. Heat transfer without medium is

A) Conduction

B) Convection

C) Radiation

D) Reflection

Answer: C

---

18. RMS velocity of gas depends on

A) Temperature

B) Mass

C) Both

D) None

Answer: C

---

19. First law of thermodynamics is

A) Energy conservation

B) Momentum conservation

C) Charge conservation

D) None

Answer: A

---

20. Efficiency of Carnot engine depends on

A) Temperature

B) Pressure

C) Volume

D) Mass

Answer: A

---

21. Frequency of wave is

A) 1/T

B) T

C) λ

D) v

Answer: A

---

22. Speed of wave is

A) fλ

B) λ/f

C) f/λ

D) None

Answer: A

---

23. Doppler effect is change in

A) Speed

B) Frequency

C) Amplitude

D) Wavelength only

Answer: B

---

24. Resonance occurs when

A) Frequency matches

B) Frequency differs

C) Amplitude zero

D) None

Answer: A

---

25. Electric field is

A) Force/charge

B) Charge/force

C) Work/charge

D) Energy

Answer: A

---

26. Unit of electric potential is

A) Volt

B) Ampere

C) Ohm

D) Watt

Answer: A

---

27. Capacitance depends on

A) Area

B) Distance

C) Medium

D) All

Answer: D

---

28. Ohmic conductor obeys

A) $V \propto I$

B) $V \propto 1/I$

C) $I \propto R$

D) None

Answer: A

---

29. Power dissipated is

A) VI

B) V^2/R

C) I^2R

D) All

Answer: D

---

30. Magnetic field due to current depends on

A) Current

B) Distance

C) Medium

D) All

Answer: D

---

31. Force on moving charge is

A) qvB

B) qE

C) qv

D) qB

Answer: A

---

32. Fleming's left hand rule is used for

A) Generator

B) Motor

C) Transformer

D) Battery

Answer: B

---

33. Induced emf depends on

A) Change in flux

B) Time

C) Both

D) None

Answer: C

---

34. Lenz's law is based on

A) Energy conservation

B) Momentum

C) Charge

D) None

Answer: A

---

35. AC current varies

A) Constant

B) Sinusoidally

C) Linearly

D) Exponentially

Answer: B

---

36. RMS value of AC is

A) $V_o/\sqrt{2}$

B) V_o

C) $\sqrt{2}V_o$

D) V_o^2

Answer: A

---

37. Power factor is

A) cosϕ

B) sinϕ

C) tanϕ

D) cotϕ

Answer: A

---

38. Total internal reflection occurs when

A) i > critical angle

B) i < critical angle

C) i = 0

D) None

Answer: A

---

39. Refractive index is

A) c/v

B) v/c

C) cv

D) c^2/v

Answer: A

---

40. Focal length of convex lens is

A) Positive

B) Negative

C) Zero

D) Infinite

Answer: A

41. Magnification is

A) Image/Object

B) Object/Image

C) Sum

D) Difference

Answer: A

---

42. Photoelectric effect proves

A) Wave nature

B) Particle nature

C) Both

D) None

Answer: B

---

43. Energy of photon is

A) hf

B) h/f

C) f/h

D) hf^2

Answer: A

---

44. Bohr radius depends on

A) n^2

B) n

C) 1/n

D) n^3

Answer: A

---

45. Binding energy is

A) Energy to break nucleus

B) Energy to form nucleus

C) Both

D) None

Answer: C

---

46. Half-life depends on

A) Mass

B) Temperature

C) Nature of nucleus

D) Pressure

Answer: C

---

47. Nuclear fission releases

A) Energy

B) Neutrons

C) Radiation

D) All

Answer: D

---

48. Semiconductor has

A) Small band gap

B) Large band gap

C) No band gap

D) Infinite

Answer: A

---

49. Diode allows current

A) One direction

B) Both

C) None

D) Reverse

Answer: A

---

50. Transistor is used as

A) Amplifier

B) Switch

C) Both

D) None

Answer: C

---

51. Logic gate AND gives

A) 1 if both 1

B) 1 if any 1

C) 0 always

D) None

Answer: A

---

52. OR gate gives

A) 1 if any 1

B) 0 always

C) 1 if both 0

D) None

Answer: A

---

53. NOT gate gives

A) Same output

B) Opposite output

C) Zero

D) One

Answer: B

---

54. De Broglie wavelength is

A) h/p

B) p/h

C) hp

D) h^2/p

Answer: A

---

55. Heisenberg principle deals with

A) Position & momentum

B) Energy

C) Mass

D) Charge

Answer: A

---

56. Laser light is

A) Coherent

B) Incoherent

C) Random

D) Diffused

Answer: A

---

57. Optical fiber works on

A) Reflection

B) Refraction

C) Total internal reflection

D) Diffraction

Answer: C

---

58. Diffraction occurs due to

A) Wave nature

B) Particle nature

C) Both

D) None

Answer: A

---

59. Interference pattern is due to

A) Superposition

B) Reflection

C) Refraction

D) Diffraction

Answer: A

---

60. Polarization proves

A) Wave nature

B) Particle nature

C) Both

D) None

Answer: A